Lecture 26.03: Linear Regression GD (Step 1 — Initialization)

📌 Topic Overview

In this lecture, Shraddha ma'am explains **Step 1 of Gradient Descent**: how to initialize weights and bias, and set learning hyperparameters.

⚙️ Hyperparameters & Initialization

1. **Weights Initialization**: Set $w = \mathbf{0}$ (vector of zeros of length d). Because linear regression has a strictly convex loss surface, zero initialization is completely safe and guaranteed to converge.
2. **Bias Initialization**: Set $b = 0.0$.
3. **Learning Rate (α)**: Step size taken along the negative gradient (e.g., 0.01 or 0.001).
4. **Number of Iterations ($epochs$)**: Total passes through the gradient update loop.

💻 Python Implementation

```
import numpy as np

def initialize_parameters(n_features):
    """
    Step 1 of Gradient Descent: Initialize parameters.
    In Linear Regression, initializing weights to zeros is safe
    because the MSE cost surface is strictly convex (parabolic).
    """
    weights = np.zeros(n_features)
    bias = 0.0
    return weights, bias

# --- Function Call & Testing ---
w_init, b_init = initialize_parameters(n_features=5)
print("Initialized Weights Vector (Zeros):", w_init)
print("Initialized Bias Scalar:           ", b_init)
```

🔑 Key Takeaways

1. Linear Regression parameters can be safely initialized to zeros.
2. Neural networks require random initialization to break symmetry, but convex linear models do not.